

Fast Food Marketing Campaign A/B Test Analysis

This is an analysis of the [Fast Food Marketing Campaign A/B Test dataset](#).

Goal of the Test

The main goal of this A/B test was to test which of the three marketing campaigns resulted in the highest total sales of a new fast-food product.

Null Hypothesis (H_0): There is NO statistically significant difference between the campaigns in terms of generated sales. In other words, they are equally successful.

Alternative Hypothesis (H_1): There is a statistically significant difference between the campaigns in terms of generated sales. In other words, one campaign performs better than the other.

For the analysis of A/B test results we will use a confidence level of 99% because, as we are comparing multiple campaigns in pairs, we suffer from the multiple testing problem.

Target Metric

Since the goal of our A/B test is to identify the most effective marketing strategy for boosting sales, the best **target metric** is **aggregated sales (*sales_in_thousands*) per campaign (*promotion*) and per location (*location_id*)**. This metric directly measures the impact of each promotion by calculating the total revenue generated at each location for each campaign. Comparing total sales across the three promotions helps determine which strategy drives the highest sales, providing clear insight into the most successful approach.

Calculations

The table below (Table 1) contains the data necessary to analyze the A/B test and reach a decision. *Note that this is only a fragment of the full data.* The SQL query can be found in the appendix.

Table 1. A fragment of the data necessary to analyze the A/B test of the Fast Food Marketing Campaign A/B test.

location_id ▼	promotion ▼	sales ▼
202	1	349.66
209	1	380.36
210	1	357.05
214	1	349.5
218	1	364.5
220	1	361.37
222	1	347.68
902	1	246.2

To assess the effectiveness of the three marketing campaigns, I performed a **One-Way ANOVA** followed by **Tukey's Honestly Significant Difference (HSD) Test** to compare total sales across the three promotional campaigns. For this, I used the [Social Science Statistics One-Way ANOVA Calculator](#). It requires entering data for each treatment group, where Treatment 1 corresponds to sales from Promotion 1, Treatment 2 to sales from Promotion 2, and Treatment 3 to sales from Promotion 3. After inputting the sales data from the respective column, the following summary was obtained:

Table 2. One-Way ANOVA Calculation Summary Details

Summary of Data						
	<i>Treatments</i>					
	1	2	3	4	5	Total
N	43	47	47			137
$\sum X$	9993.03	8897.93	10408.52			29299.48
Mean	232.396	189.3177	221.4579			213.865
$\sum X^2$	2494980.1033	1839217.3349	2502613.9494			6836811.3876
Std.Dev.	64.1129	57.9884	65.5355			64.7781

1. The results of the One-Way ANOVA test:

Table 3. One-Way ANOVA Test Result Details

Result Details				
Source	SS	df	MS	
Between-treatments	45796.6972	2	22898.3486	$F = 5.84579$
Within-treatments	524886.7469	134	3917.0653	
Total	570683.444	136		

The F -ratio value is 5.84579. The p -value is .003681. The result is significant at $p < .01$.

The **F-Statistic** value of ~5.85 represents a great difference between group means.

Since the p -value (~0.004) is less than the significance threshold of 0.01, we **reject the null hypothesis** and conclude that **there is a significant difference between the campaigns in terms of generated sales**.

2. Tukey's HSD Test:

To identify which **specific** promotional campaigns differ from each other, Tukey's HSD test was performed. The **pairwise comparisons** are as follows:

Table 4. Tukey's HSD Test Result Details

Pairwise Comparisons		HSD _{.05} = 31.0690 HSD _{.01} = 38.8537	Q _{.05} = 3.3517 Q _{.01} = 4.1915
T ₁ :T ₂	M ₁ = 232.40 M ₂ = 189.32	43.08	$Q = 4.65$ ($p = .00369$)
T ₁ :T ₃	M ₁ = 232.40 M ₃ = 221.46	10.94	$Q = 1.18$ ($p = .68238$)
T ₂ :T ₃	M ₂ = 189.32 M ₃ = 221.46	32.14	$Q = 3.47$ ($p = .04081$)

T₁ vs. T₂ (Promotion 1 vs. Promotion 2)

- Promotion 1 **significantly** outperforms Promotion 2 in average sales, with a difference of 43.08.
- The p-value (~ 0.004) is below the significance threshold $\alpha = 0.01$, making this result **statistically significant at the 99% confidence level**.
- The low p-value suggests that the likelihood of observing this difference (or a larger one) purely by chance is very small.

T₁ vs. T₃ (Promotion 1 vs. Promotion 3)

- There is **no statistically significant difference** between Promotion 1 and Promotion 3 in terms of average sales.
- The p-value (~ 0.682) is well above $\alpha = 0.01$, indicating **no strong evidence that one outperforms the other**.

T₂ vs. T₃ (Promotion 2 vs. Promotion 3)

- Promotion 3 **slightly outperforms** Promotion 2, with a difference of 32.14 in average sales.
- The p-value (~ 0.041) falls below the 0.05 significance threshold but not below 0.01, meaning the result is **statistically significant at the 95% confidence level but not at 99%**.
- This suggests that, **under the 99% confidence level**, there is **insufficient evidence** to confidently conclude that Promotion 3 outperforms Promotion 2.

Conclusion:

- **Promotion 1 significantly outperforms Promotion 2** in total sales with strong evidence at the 99% confidence level.
- There is **no significant difference** between **Promotion 1** and **Promotion 3** in total sales, indicating that both promotions perform similarly.
- **Promotion 3 slightly outperforms Promotion 2**, but the result is only significant at the 95% confidence level, not at 99%. This indicates that the performance difference isn't strong enough to conclusively favor Promotion 3 at the 99% confidence level.

Decision:

Promotion 1 is significantly more effective than Promotion 2, but Promotion 3 performs similarly to both (at a confidence level of 99%) creating some uncertainty. It remains unclear whether Promotion 1 is truly the best or if Promotion 1 and 3 are equally effective. Revisiting the test design is recommended to clarify the results.

Appendix

Query for table 1

```
SELECT
location_id,
promotion,
ROUND(SUM(sales_in_thousands), 2) AS sales
FROM `tc-da-1.turing_data_analytics.wa_marketing_campaign`
GROUP BY location_id,
promotion
ORDER BY promotion
```

Sample Ratio Mismatch

I noticed that the number of locations differs among the promotion variants, with Promotion 1 having four fewer locations than the other promotions. This discrepancy represents approximately 2.92%. If the locations were properly randomized, we would expect the proportions to be as close to 33.3% as possible. However, a perfectly equal distribution is not possible (137 locations / 3 = 45.6 stores). Therefore, the best we could do is allocate 45, 46, and 46 stores across the campaigns. **The question is: is the difference statistically significant?**

Promotion	Location_count	Proportion	Expected_location_count
1	43	31.39%	46
2	47	34.31%	46
3	47	34.31%	45

To determine whether the difference in the number of locations across variants is statistically significant, we can perform a Chi-square goodness-of-fit test. This test assesses whether the observed distribution of locations deviates significantly from the expected equal distribution (i.e., 33.3% for each variant).

- **Null Hypothesis (H_0):** There is NO significant difference in the distribution of locations across the three marketing campaigns.
- **Alternative Hypothesis (H_1):** There is a significant difference in the distribution of locations across the three marketing campaigns.

Key Results (calculated using [GraphPad Chi-square test](#)):

Chi-Square Statistic (χ^2) = 0.306

Degrees of Freedom (df) = 2

P-value = 0.858

Interpretation:

The p-value (0.858) is much higher than typical significance levels (e.g., 0.05 or 0.01), meaning that the difference in proportions between the three groups is **not statistically significant**.

This means that the observed differences in the number of locations (43 vs. 47 vs. 47) are likely **due to random chance** and **do not provide strong evidence against the null hypothesis**.

Conclusion:

Since the p-value (0.858) is greater than the common significance thresholds (0.05 or 0.01), **I fail to reject the null hypothesis**. This indicates that the observed difference in proportions is not statistically significant, and we can conclude that the proportions of locations in each category are very similar, within the expected range of variation.